Question 47

05: Improve Developer Productivity

What is the recommended approach for modernizing/refactoring legacy code (such as translating COBOL or Python 2 to modern language standards) using GitHub Copilot?

A. Prompt Copilot to rewrite the entire application architecture in one request.

✔ **B. Refactor code incrementally function by function, using tests to verify functional equivalence at each step.**

C. Delete existing legacy files and let Copilot generate the replacement code from scratch without context.

D. Disable prompt filters before converting legacy syntax.

Correct Answer: B

Refactoring complex or legacy code succeeds best when broken down into modular, incremental steps. Converting function by function while running unit tests ensures behavioral parity and prevents subtle translation bugs.

Option Breakdown

- **A. Rewrite entire app at once:** Incorrect. Requesting full codebase rewrites exceeds context windows and increases bug risks.
- **B. Incremental refactoring with test verification:** Correct. Best practice approach for preserving functional equivalence.
- **C. Delete files and generate from scratch:** Incorrect. Removing context degrades suggestion quality and risks losing domain rules.
- **D. Disable prompt filters:** Incorrect. Security filters protect code quality and cannot be bypassed.

💡 **GH-300 Exam Tip**

Pair `/explain` to understand complex legacy algorithms with `/tests` to capture existing behavior before initiating incremental refactoring requests.